

Rapid Prototyping Camp (RPC)

Abstract

Project Title: AI-powered Digital Platform for Cognitive Screening (DPCS)

Team Name: Deep Cognition

Category/Track: HealthTech, AIML

I. The Problem (The Why/Goal)

Goal: Early cognitive decline and **dementia** frequently go undiagnosed due to limited access to standard screening, resource constraints, and highly subjective, manual evaluations. Paper and in-person tests do not scale, and results may be affected by environmental inconsistencies. The current AI-based and web-based solutions are very costly. Our solution aims to provide an accurate open-source alternative that is easy to use and easy to adopt by doctors. Starting with **screening tests for early-stage dementia** and later developing for other cognitive diseases as well.

Target User: Healthcare providers, neuropsychologists, and older adults—especially those in remote or underprivileged regions—who need scalable, accessible, and affordable cognitive assessment solutions that provide accurate results.

II. The Solution (The What/Concept/Key Features)

Concept: Dementia Project is a next-generation browser platform delivering **digitized versions of leading cognitive tests**, with immediate plans to aggregate multiple scientifically proven assessments. Ultimately, we will build a unified “super-test” combining the strengths of diverse tools for robust, AI-assisted early detection and precision scoring.

Technology & Key Features:

- **Next.js** 14 frontend with responsive, interactive test modules (canvas input, audio, MCQ)
- **Fast API** backend (**Python**) with AI stubs for vision and voice analysis; **MongoDB Atlas** (AES-256 encrypted) for secure data storage
- Automated real-time scoring with >95% agreement to clinical standards
- Idle/attention detection, **user time tracking**, and environment metadata capture (with ongoing integration of eye tracking and behaviour analysis for deeper pattern identification)
- **Highly accessible UX:** adaptive layouts, large-font, high-contrast, full mobile/desktop support
- Remote **self-administration** and exportable, structured clinical reports

III. Impact and Innovation (The So What)

Innovation: Merges diverse digital cognitive assessments in a single, extensible platform with deterministic scoring algorithms. Continuous integration of additional validated tests, advanced environmental/context capture, and **AI-powered eye and speech** (tone, hesitation, vocabulary, or recall) feature extraction, **vernacular languages** (like English, Hindi, Marathi, etc.) represent next-gen innovation for real-world diagnosis and research.

Project Status: Currently developed a working prototype that contains the **full interactive MoCA test**. Expanding to more assessment modules, environment metrics (e.g., **attention/eye-tracking**, **precise test time tracking**), and longitudinal data capture in future MVP. Solution demonstrates robust tech stack and accurate scoring pipeline.

Prototype Link: <https://dpcs-722m.vercel.app/>

Future Potential:

- Full library of digital cognitive tests, plus a “super-assessment” synthesizing best-in-class metrics
- Advanced features: Real AI vision, speech assessment (detecting subtle voice/speech issues), deep learning for behavioural/interaction analytics, digital phenotype mapping
- EHR/FHIR integration for clinical adoption
- Multilingual support, tablet/native app
- Scalable for population screening, clinical trials, and tele-neurology

IV. Implementing Technology and Methodology Used

Core Stack:

- Frontend: Next.js + TypeScript, Tailwind, Konva.js/React Canvas Draw
- Backend: Python Fast API, Motor (MongoDB)
- Database: MongoDB Atlas (AES-256 at rest)
- Authentication: NextAuth.js (OAuth/email)
- Deployment: Vercel (frontend), Render (Backend)
- Security: AES-256 encryption, JWT, HTTPS

Architecture: RESTful APIs handle each test and scoring module; session-based data and results.

Methodology:

- Deterministic and heuristic ML-scoring (vision, language/speech) with fuzzy matching for text and audio
- Automated/manual-review triggers for results below confidence thresholds
- Upcoming voice models to analyse speech features and identify speech impairment patterns, as several tests incorporate spoken responses and recordings

- **Time/behaviour tracking, eye-tracking** slated for next releases

V. Datasets Used

Primary Data: Platform collects structured digital test interactions (drawing, MCQ, speech/audio, geolocation, etc) per user session.

Preprocessing: Captures raw user signals—draws converted to vectors, speech transcribed and analysed, scoring via fuzzy-logic, heuristics, and ML wherever validated.

Custom Data: All test data generated during user interaction. Real clinical samples or standardized patient data may be incorporated for future model training (obeying data privacy laws).

Reference Links & Supporting Research

We researched from:

- Cleveland Clinic, Alzheimer's Association, WHO, Columbia University

- Meta-analysis of MoCA in online and remote formats:

<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2024.1369766/full>

<https://www.tandfonline.com/doi/full/10.1080/13854046.2024.2397835>

- Digital cognitive screening tool validation:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10680059/>

<https://www.jmir.org/2025/1/e66735>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12267269/>

<https://aging.jmir.org/2025/1/e65292>

<https://www.sciencedirect.com/science/article/abs/pii/S1568163721002531>

- AI/ML for voice and speech-based cognitive assessment:

<https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0325177>

<https://arxiv.org/html/2502.08862v1>

<https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2024.1342907/full>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4876915/>

<https://www.sciencedirect.com/science/article/pii/S2772598724000473>

<https://www.sciencedirect.com/science/article/pii/S0263224125021037>

- Official MoCA resources:

<https://mocacognition.com>